Advanced Engineering Mathematics

Integral Transform Methods

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1 Integral Transform Methods

Key Concept

An integral transform converts a function from one domain into another domain by integrating it against a kernel:

$$F(\lambda) = \int_a^b K(\lambda, t) f(t)\, dt.$$

Here, $f(t)$ is the original function, $K(\lambda, t)$ is the kernel, and $F(\lambda)$ is the transformed function.

Integral transform methods are powerful because they replace difficult operations in the original domain with simpler operations in the transform domain. In engineering mathematics, the most common use is to convert differential equations, convolution problems, and signal-processing problems into algebraic or spectral forms.

The central idea is:

$$\text{difficult problem in } t \quad \longrightarrow \quad \text{simpler problem in transform variable} \quad \longrightarrow \quad \text{inverse transform.}$$

Engineering Note

In engineering, transform methods are used in circuit analysis, control systems, signal processing, vibration analysis, communication systems, image processing, and numerical solution of differential equations.

General Workflow

$$\boxed{f(t) \xrightarrow{\text{transform}} F(\lambda) \xrightarrow{\text{solve or analyze}} \text{simpler expression} \xrightarrow{\text{inverse transform}} f(t)}$$

Key Concept

The four transform methods in this lecture are:

Laplace Transform	for causal systems and initial-value problems,
Fourier Transform	for frequency analysis of aperiodic signals,
Hilbert Transform	for phase, analytic signals, and spectral relationships,
Z Transform	for discrete-time sequences and digital systems.

Connection to Previous Topics

Fourier series expanded periodic functions as sums of sine and cosine waves. Integral transforms extend this idea to broader classes of functions.

Fourier series handles periodic signals, while the Fourier transform handles aperiodic signals. The Laplace transform extends this further by adding exponential damping through e^{-st}, making it especially useful for functions defined for $t > 0$.

Engineering Note

A useful way to remember the difference is:

$$\begin{aligned}
\text{Fourier Series} &: \text{ periodic signal } \to \text{ discrete frequencies},\\
\text{Fourier Transform} &: \text{ aperiodic signal } \to \text{ continuous frequencies},\\
\text{Laplace Transform} &: \text{ causal signal } \to \text{ complex-frequency model},\\
\text{Z Transform} &: \text{ discrete signal } \to \text{ complex discrete-frequency model}.
\end{aligned}$$

Example

Example: Why transforms are useful

Suppose an engineering system is modeled by

$$y''(t) + 3y'(t) + 2y(t) = x(t).$$

In the time domain, this is a differential equation. After applying a transform, derivatives become algebraic factors. For example, under the Laplace transform,

$$L[y'(t)] = sY(s) - y(0),$$

and

$$L[y''(t)] = s^2Y(s) - sy(0) - y'(0).$$

Thus, the differential equation becomes an algebraic equation in $Y(s)$:

$$\left(s^2Y(s) - sy(0) - y'(0)\right) + 3\left(sY(s) - y(0)\right) + 2Y(s) = X(s).$$

Solving for $Y(s)$ is often easier than solving directly for $y(t)$.

Common Mistakes

Key Concept

Common mistakes in transform methods include:

1. Forgetting the domain of the transform variable.
2. Ignoring convergence conditions.
3. Applying inverse transforms without checking the form.
4. Confusing continuous-time transforms with discrete-time transforms.
5. Treating transform tables as formulas without understanding assumptions.

Engineering Note

The transform is not just a computational shortcut. It changes the viewpoint of a problem. In engineering, this is often the difference between analyzing a signal in time and analyzing the same signal in frequency, stability, or pole-zero form.

1.1 Laplace Transform

Motivation and Intuition

The Laplace transform is designed for functions that begin at $t = 0$ and are usually taken to be zero for $t < 0$. This makes it natural for engineering systems that start operating at a particular time, such as circuits after a switch is closed, mechanical systems after a force is applied, or control systems after an input signal begins.

The main advantage is that it converts differentiation into multiplication by s, while also carrying initial conditions into the transformed equation.

Key Concept

For a function $f(t)$ such that $f(t) = 0$ for $t < 0$, the Laplace transform is defined by

$$L[f(t)] = F(s) = \int_0^\infty f(t)e^{-st}\,dt.$$

The inverse transform is written as

$$L^{-1}[F(s)] = f(t).$$

The factor e^{-st} acts like a damping or weighting factor. If $f(t)$ grows too quickly, the integral may not converge. Therefore, functions are usually required to be piecewise continuous and of exponential order.

Key Concept

A function $f(t)$ is of exponential order if there exist constants M and k such that

$$|f(t)| \leq Me^{kt}, \qquad t > 0.$$

Laplace Transforms of Common Functions

Key Concept

The following table summarizes commonly used Laplace transforms. These are especially useful when solving initial-value problems and circuit/system models.

Function $f(t),\ t \geq 0$	**Laplace Transform** $F(s) = L[f(t)]$
1	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
e^{at}	$\frac{1}{s-a}$
t	$\frac{1}{s^2}$
$t^n,\ n = 1, 2, 3, \ldots$	$\frac{n!}{s^{n+1}}$
$\sin(at)$	$\frac{a}{s^2+a^2}$
$\cos(at)$	$\frac{s}{s^2+a^2}$
$\sinh(at)$	$\frac{a}{s^2-a^2}$
$\cosh(at)$	$\frac{s}{s^2-a^2}$
$H(t-a)$	$\frac{e^{-as}}{s}$
$\delta(t-a)$	e^{-as}

Engineering Note

The Laplace transform table is most useful when the function is causal, meaning the system response begins at $t = 0$. In engineering, this matches switched circuits, forced mechanical systems, and control-system inputs.

Key Properties

Key Concept

Linearity

$$L[c_1 f(t) + c_2 g(t)] = c_1 L[f(t)] + c_2 L[g(t)].$$

Key Concept

Derivative Property If $F(s) = L[f(t)]$, then

$$L[f'(t)] = sF(s) - f(0).$$

For the second derivative,

$$L[f''(t)] = s^2 F(s) - sf(0) - f'(0).$$

This property is the reason Laplace transforms are especially effective for initial-value problems.

Example

Example 1: Find the Laplace transform of $f(t) = 3e^{2t} - 4\sin(5t)$.

Using linearity,

$$L[3e^{2t} - 4\sin(5t)] = 3L[e^{2t}] - 4L[\sin(5t)].$$

From the basic transform formulas,

$$L[e^{2t}] = \frac{1}{s-2},$$

and

$$L[\sin(5t)] = \frac{5}{s^2+25}.$$

Therefore,

$$F(s) = \frac{3}{s-2} - \frac{20}{s^2+25}.$$

Example

Example 2: Find $L^{-1}\left[\frac{s+2}{s^2+1}\right]$.

Separate the expression:

$$\frac{s+2}{s^2+1} = \frac{s}{s^2+1} + \frac{2}{s^2+1}.$$

Using the table,

$$L[\cos t] = \frac{s}{s^2+1},$$

and

$$L[\sin t] = \frac{1}{s^2+1}.$$

Thus,

$$L^{-1}\left[\frac{s}{s^2+1}\right] = \cos t,$$

and

$$L^{-1}\left[\frac{2}{s^2+1}\right] = 2\sin t.$$

Therefore,

$$f(t) = \cos t + 2\sin t.$$

Example

Example 3: Solve the initial-value problem

$$y'(t) + 2y(t) = e^{-t}, \qquad y(0) = 3.$$

Apply the Laplace transform:

$$L[y'(t)] + 2L[y(t)] = L[e^{-t}].$$

Using

$$L[y'(t)] = sY(s) - y(0),$$

we get

$$sY(s) - 3 + 2Y(s) = \frac{1}{s+1}.$$

Combine the $Y(s)$ terms:

$$(s+2)Y(s) - 3 = \frac{1}{s+1}.$$

Solve for $Y(s)$:

$$(s+2)Y(s) = 3 + \frac{1}{s+1}.$$

$$Y(s) = \frac{3}{s+2} + \frac{1}{(s+1)(s+2)}.$$

Use partial fractions:

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}.$$

Multiplying by $(s+1)(s+2)$ gives

$$1 = A(s+2) + B(s+1).$$

Set $s = -1$:

$$1 = A(1),$$

so

$$A = 1.$$

Set $s = -2$:

$$1 = B(-1),$$

so

$$B = -1.$$

Therefore,

$$Y(s) = \frac{3}{s+2} + \frac{1}{s+1} - \frac{1}{s+2}.$$

$$Y(s) = \frac{1}{s+1} + \frac{2}{s+2}.$$

Taking the inverse Laplace transform,

$$y(t) = e^{-t} + 2e^{-2t}.$$

Engineering Insight

Engineering Note

In circuit analysis, capacitors and inductors produce differential equations. The Laplace transform converts these equations into algebraic equations involving impedances. For example, differentiation in time corresponds to multiplication by s, allowing engineers to analyze transient and steady-state behavior in one framework.

Common Mistakes

Key Concept

Common mistakes in the Laplace transform include:

1. Forgetting initial conditions in $L[y'(t)]$ and $L[y''(t)]$.
2. Writing $L[e^{at}] = \frac{1}{s+a}$ instead of $\frac{1}{s-a}$.
3. Applying transform formulas without checking convergence.
4. Forgetting to decompose rational functions before inversion.